DENOG16 **IXP Update**

Tim Kleefass

IXP Update

- As part of the annual DENOG conference, we would like to give all IXPs in Germany the opportunity to provide a condensed update to the community

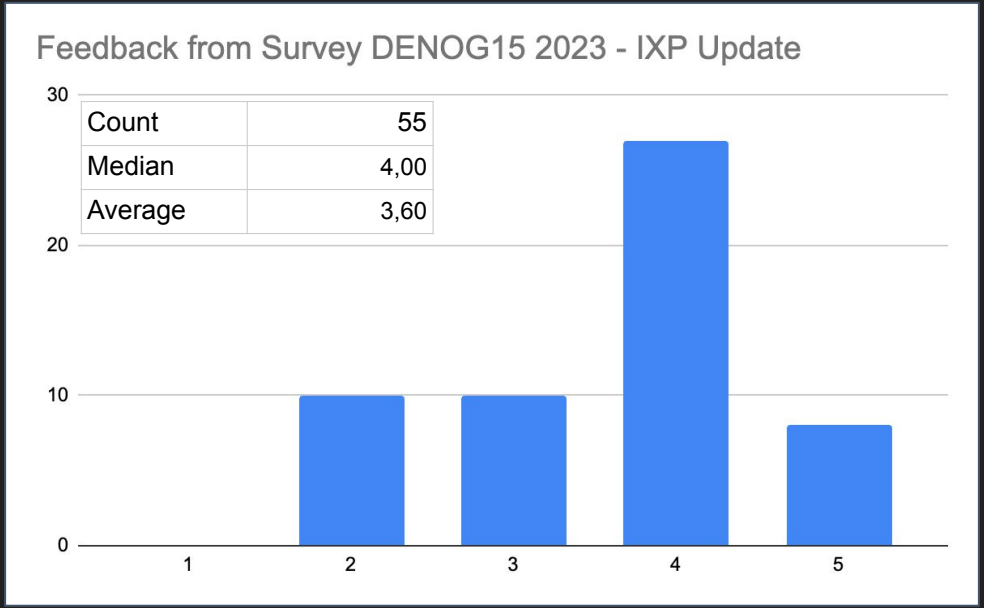
- We asked 21 IXP operators
  - 15 responded with 24 Metros
  - Last year: 12 responses with 16 Metros

Feedback from Survey DENOG15 2023

Started to early

Just an iteration of boring numbers / pretty boring

I would scrap that - could be an email



IXP's highlight of the year are most interesting

stats where interesting, no major adjustments

Missed comparison to last year

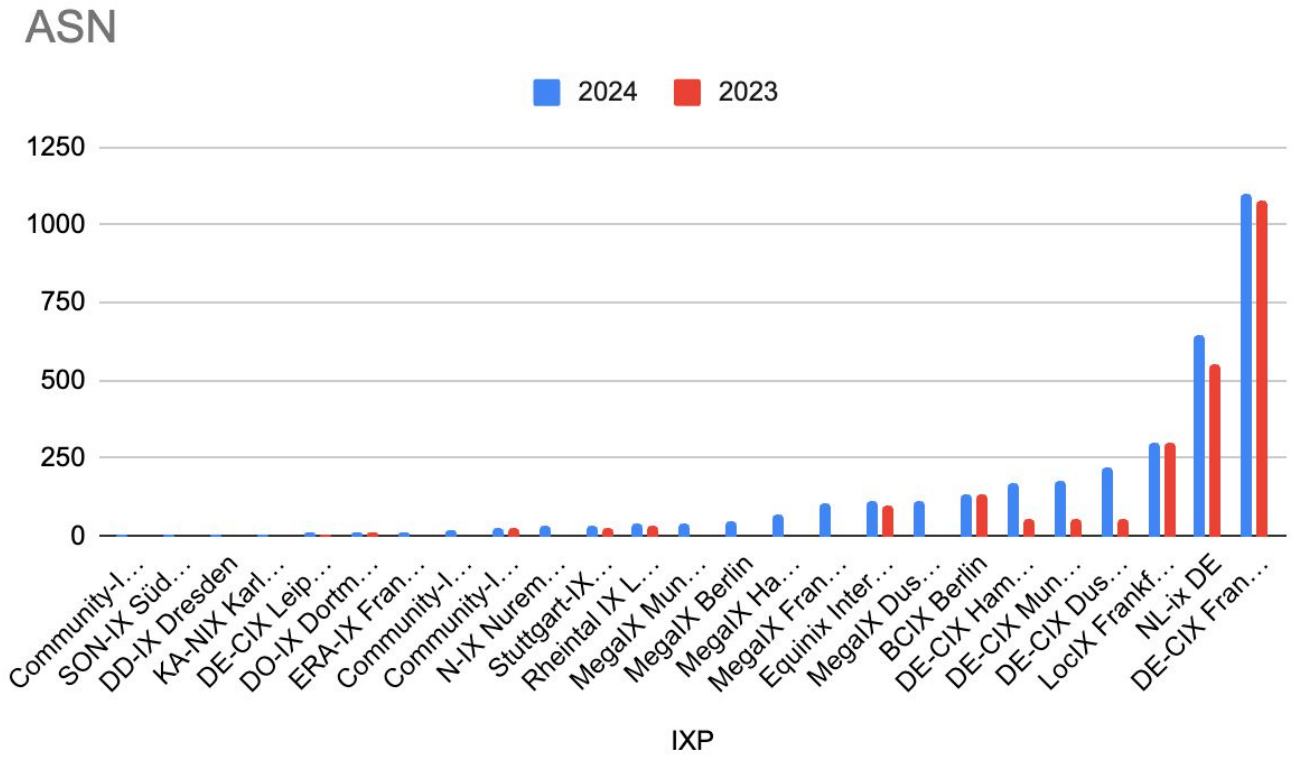
Interesting statistic could be: hardware/vendor in use

Aggregated view on Features

Number of ASNs (per metro)

Average: 138
Last year: 183

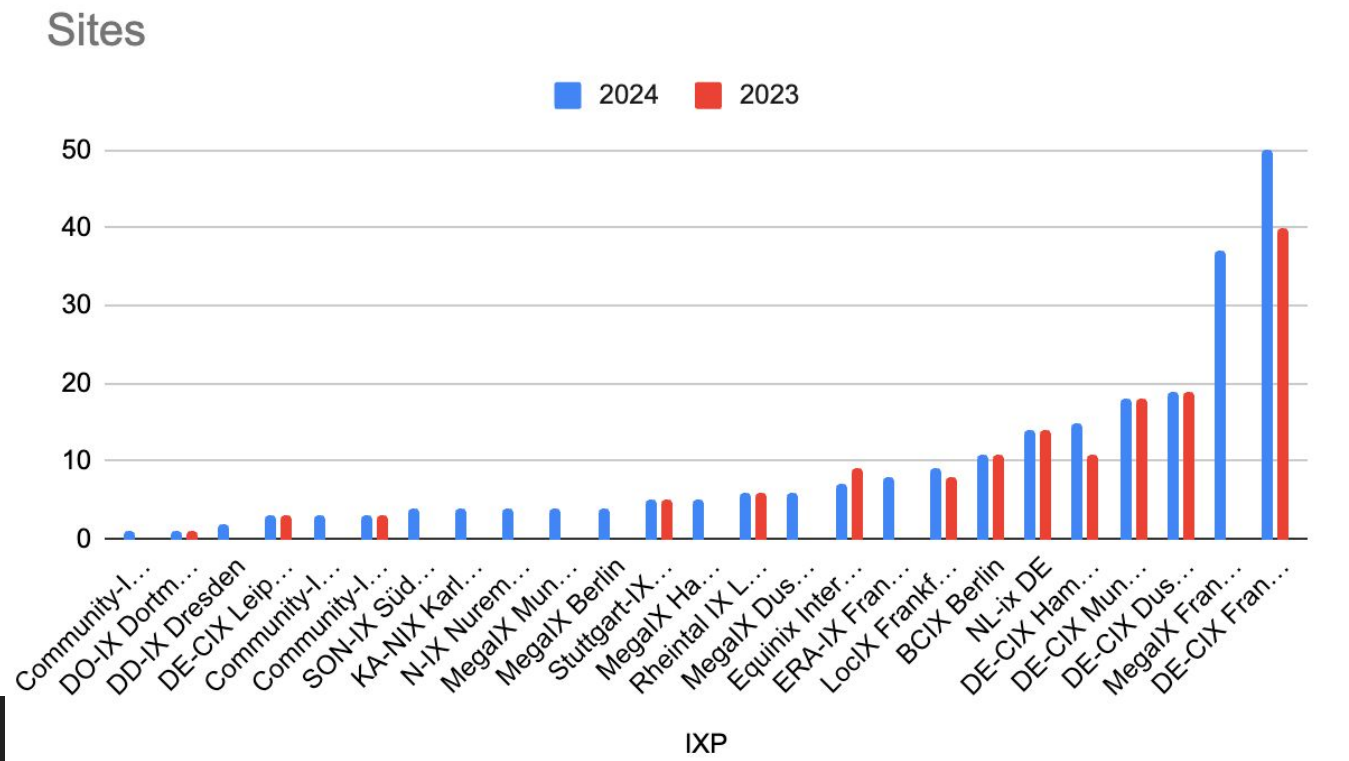
	2024	2023
Avg	138	183
Max	1098	1080
Min	4	0



Number of sites (per metro)

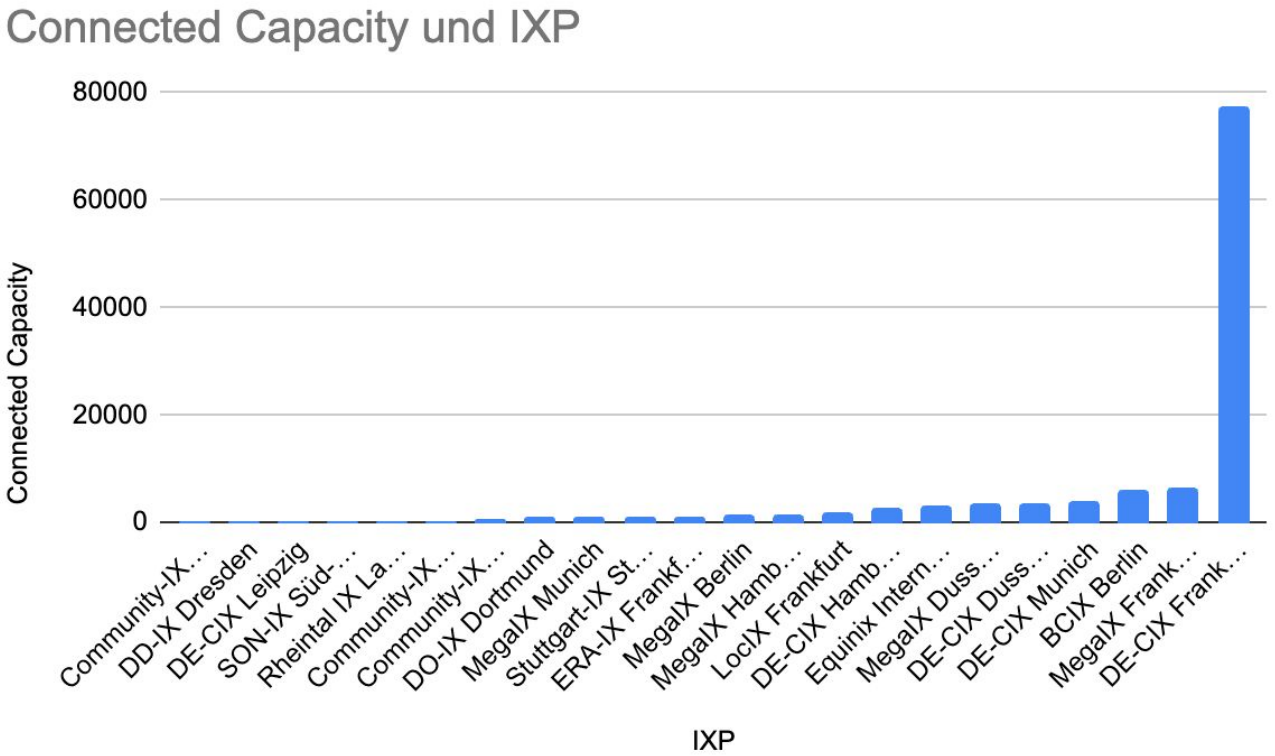
Average: 9,7
Last year: 10,4

	2024	2023
Avg	9,7	10,4
Max	50	40
Min	1	0



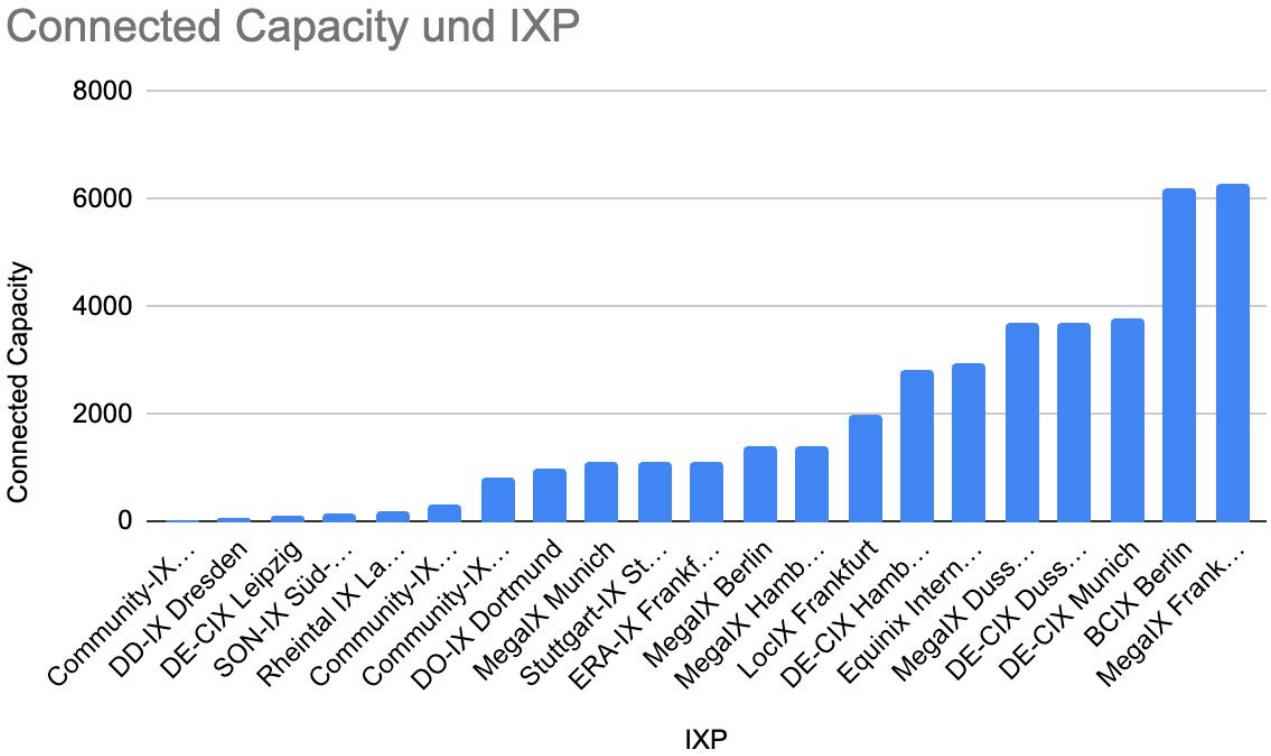
Connected Capacity

Average: 5.353G
(no useful last year data)



Connected Capacity (without DE-CIX FRA)

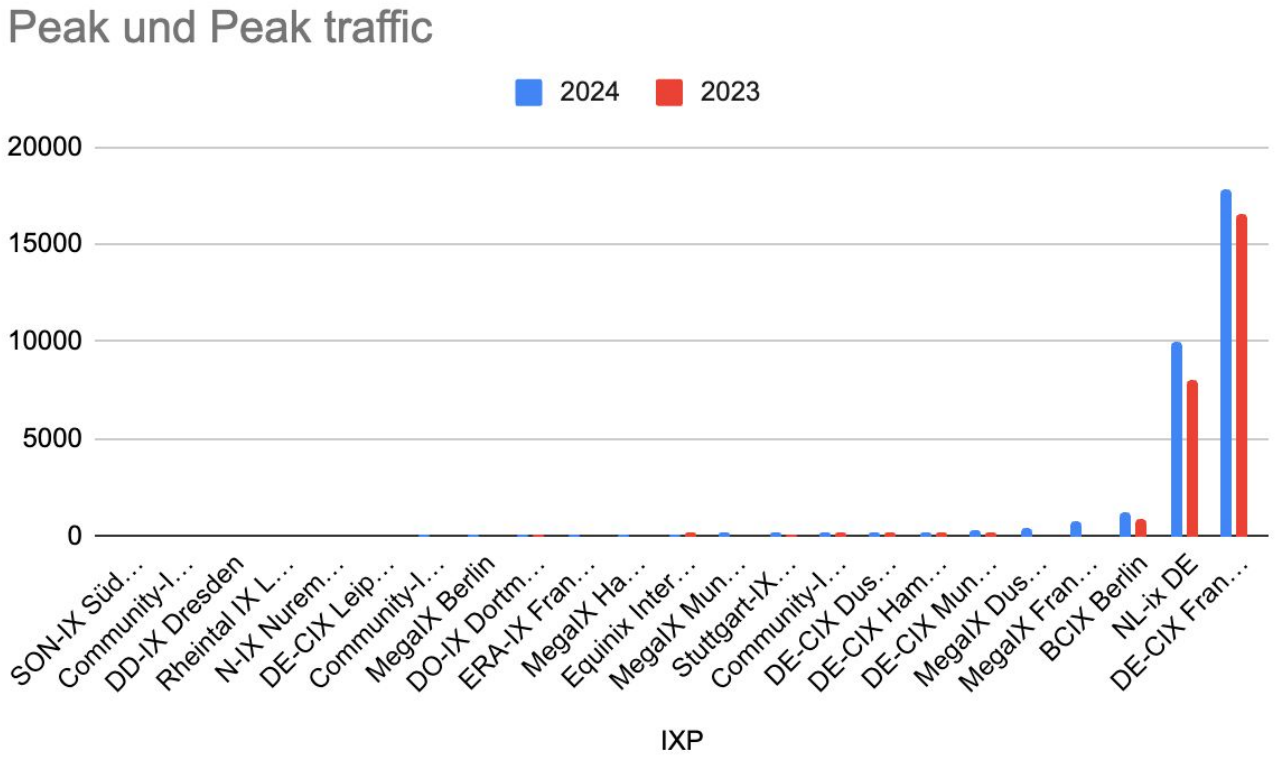
Average: 1.917G
(no useful last year data)



Peak traffic

Average: 1.390G
Last year: 1.778G

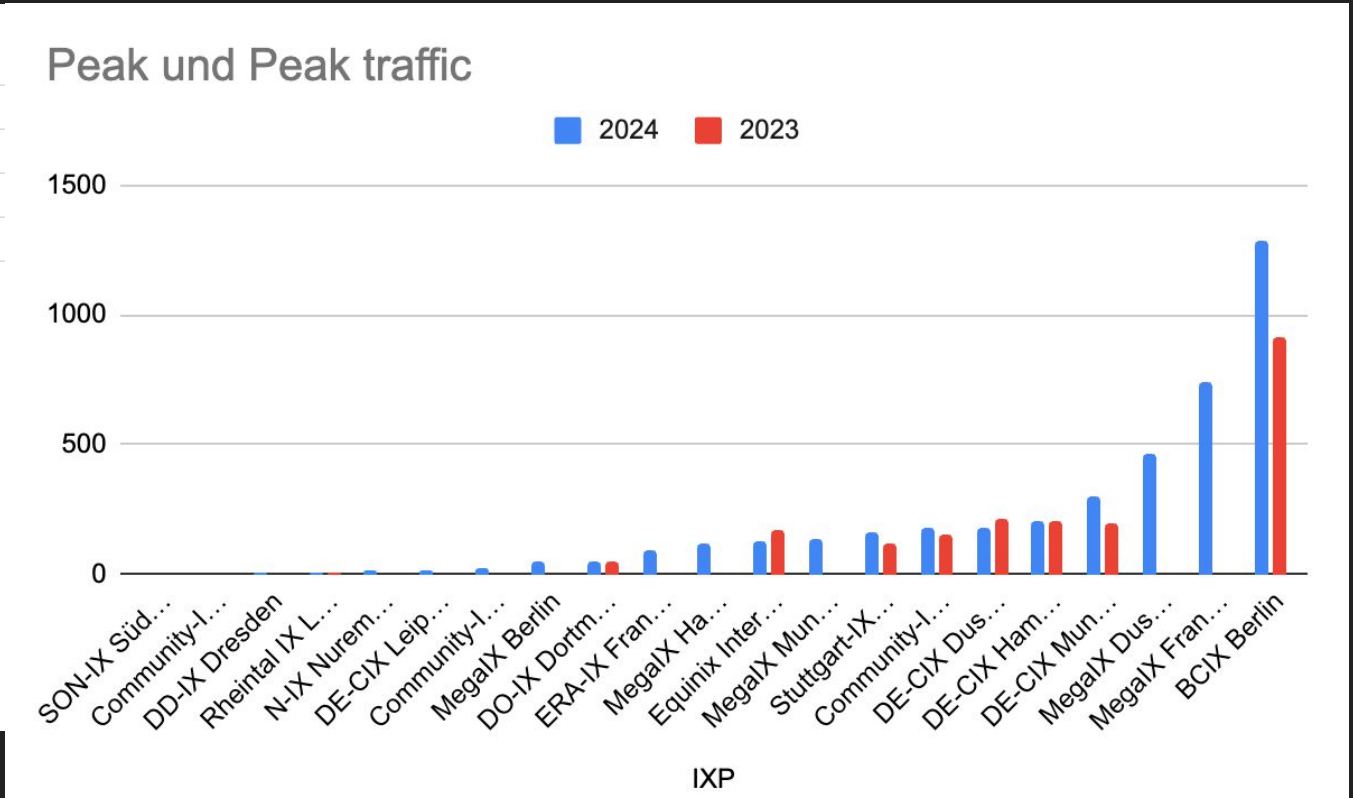
	2024	2023
Avg	1.390	1.778
Max	17.800	16.620
Min	0,1	0



Peak traffic (without DE-CIX FRA and NL-IX)

Average: 197G

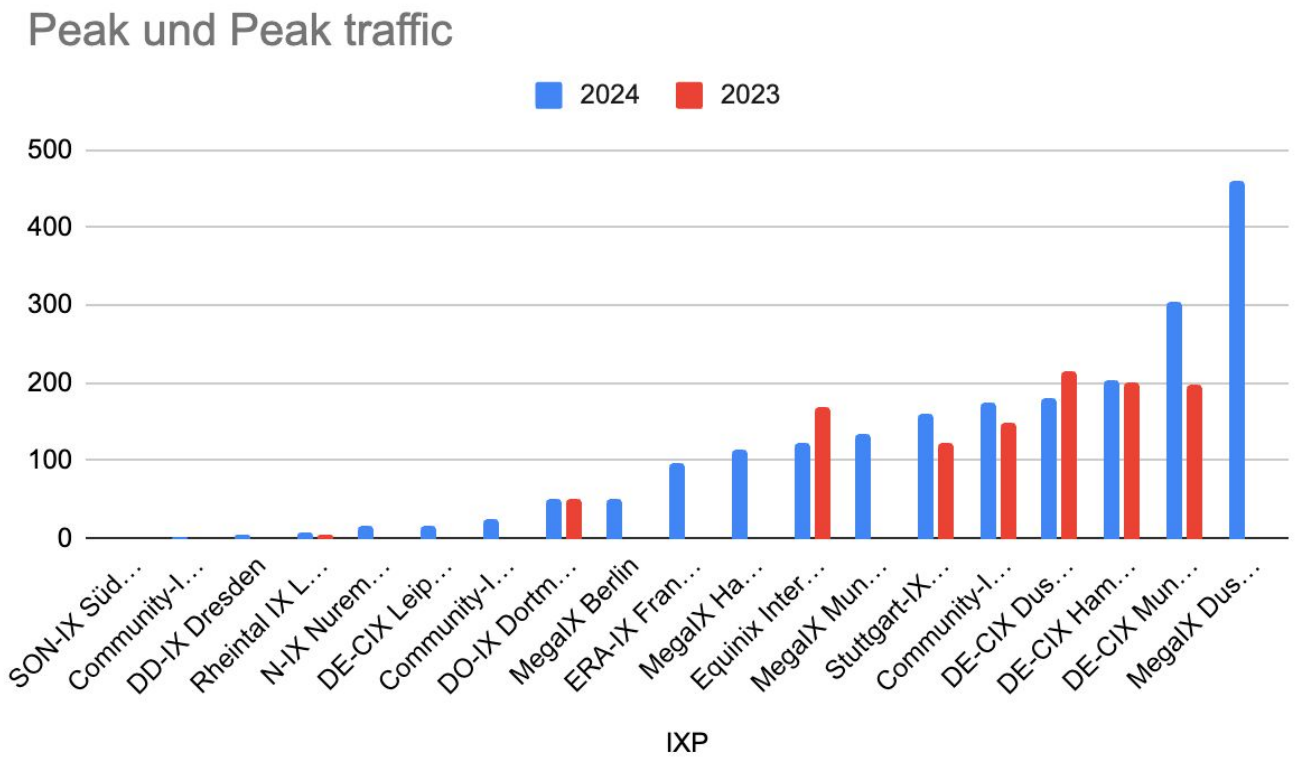
	2024	w/o Top 2
Avg	1.390	197
Max	17.800	1.285
Min	0,1	0,1



Peak traffic (and w/o BCIX, MegaIX Frankfurt)

Average: 197G

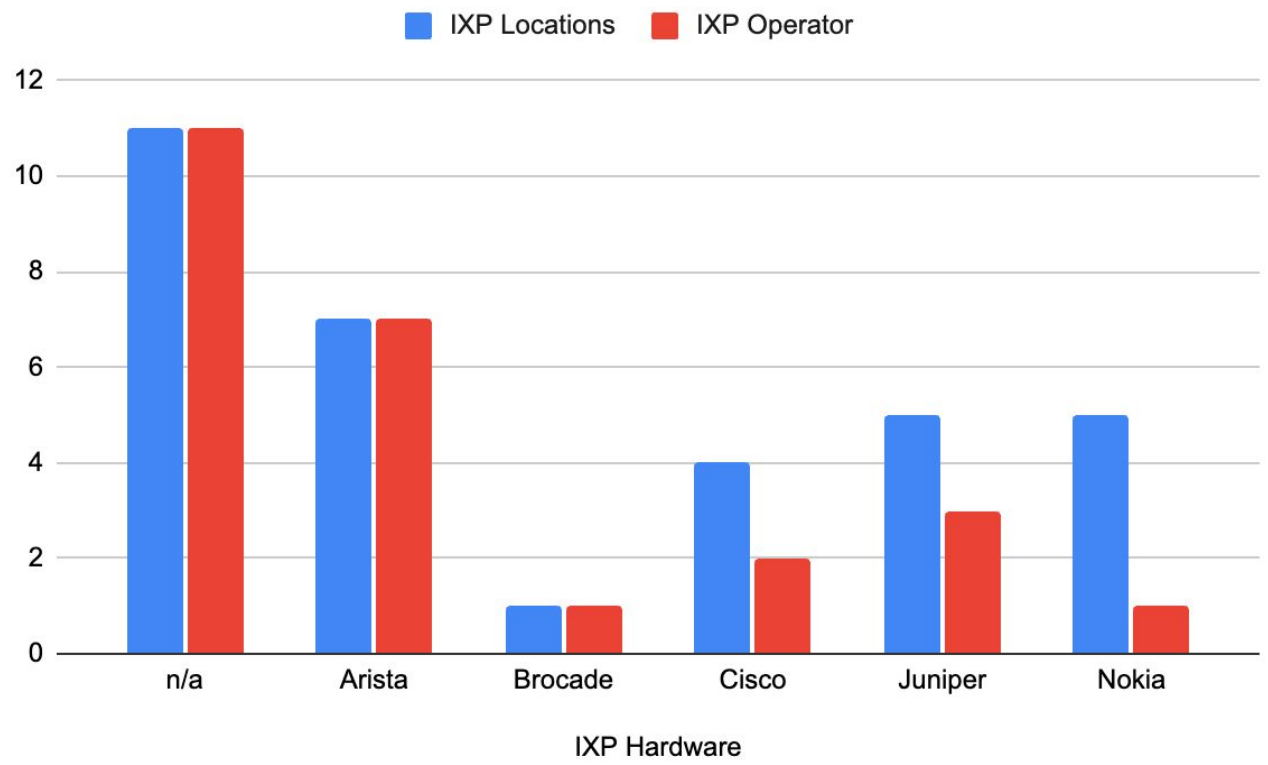
	2024	w/o Top 4
Avg	1.390	112
Max	17.800	460
Min	0,1	0,1



Hardware

Also mentioned

- FlexOptix
- Smartoptics
- Infinera
- Dell, HPE, IBM



Sustainability

Sustainability

Answers:

- We used the most efficient equipment, we currently push in total more than 1 Tbit/s while only using below <2 kW of power. We deploy the same stack in Frankfurt metro, so we expect similar efficiency, especially if we upgrade to our new 400G stack.

ERA-IX

- We strive to identify and replace power-hungry hardware and activity.

BCIX

IXP News / new Features

Google is leaving the route servers



Bill Woodcock • 3rd+
Executive Director at Packet Clearing House
7mo •

+ Follow ···

Google takes a big step backward in their connectivity to the rest of the Internet:

As part of Google's effort to simplify peering operations and increase routing security we have decided to stop advertising and receiving prefix information from Route-Servers in Internet Exchanges. The BGP session with the route servers will remain established. We are planning to implement the change in your IX in the next few weeks.

We won't disconnect from the IX and we will keep current BGP bilateral sessions with individual peers on the IX. We invite other networks that rely on Route-Servers to request a bilateral session. Bilateral session request can be done at: https://isp.google.com/iwantpeering

We kindly request you to inform peers in your Internet Exchange of these future changes so they can request bilateral sessions with us if they desire to do so.

Source: LinkedIn

Content Distribution Network

Answers:

- (Anonymously) Large content networks move a step back from plain peering in general which results in disconnecting smaller / regional IXPs and removing BGP sessions from route servers. At the same time these large content networks get bigger and bigger (concentration is moving on) and push more and more traffic.
- How do we make sure small-to-medium sized networks can have premium access to these large content networks?
- Obviously, the large content networks also provide an answer to this: Google VPP (Verified Peering Provider) and MAPS (Microsoft Azure Peering Service).

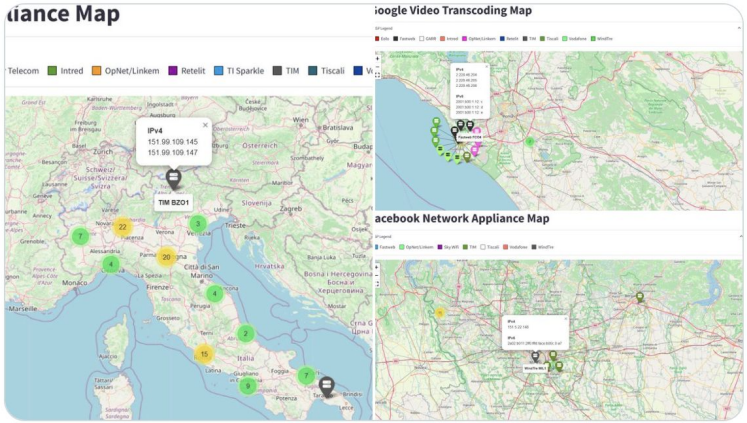
cache.amennillo.eu

- It would be great if sites like https://cache.amennillo.eu/ would exist for other (as in the German) market.

Just updated the data of cache.amennillo.eu , where I try to map the "appliances" of some CDNs hosted by Italian ISPs

Reports of errors and /o omissions are welcome

Was this translation accurate? Give us feedback so we can improve:



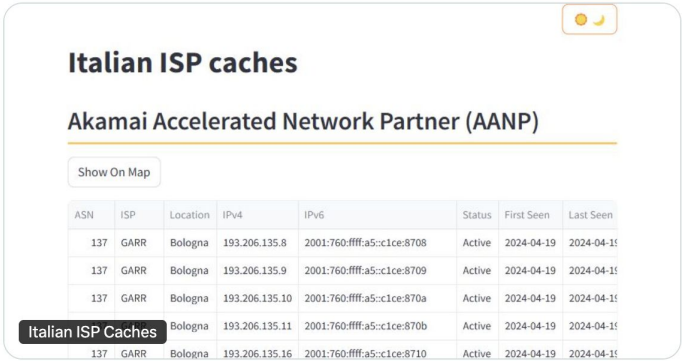
9:46 PM · Oct 29, 2024 · **208** Views

Flavio Luciani
@flavioluciani81

Among other #CDN, you can find the list of #DAZN edges available within Italian operators here: cache.amennillo.eu/#dazn-edge

Thanks @handymenny for that.

#livestreaming



From cache.amennillo.eu

11:17 AM · Oct 30, 2024 · **562** Views

https://x.com/handymenny/status/1851365114181509622

Autonomous System Provider Authorization (ASPA)

“Autonomous System Provider Authorization (ASPA) objects are created and distributed the same way as Route Origin Authorizations (ROAs). While ROAs state which ASNs are authorized to announce given prefixes, ASPAs state which ASNs are allowed to propagate their routes.”
https://manrs.org/2023/02/unpacking-the-first-route-leak-prevented-by-aspa/

Estimating the Timeline for ASPA Deployment
https://manrs.org/2023/05/estimating-the-timeline-for-aspa-deployment/

“This document defines the semantics of an Autonomous System Provider Authorization object in the Resource Public Key Infrastructure to verify the AS_PATH attribute of routes advertised in the Border Gateway Protocol.”
https://datatracker.ietf.org/doc/html/draft-ietf-sidrops-aspa-verification-04

MANRS for IXPs

The MANRS IXP Program outlines the key steps IXPs can take to demonstrate their commitment to routing security and bring significant improvement to the resilience and security of their peering relationships.

Participation in the program provides an opportunity for IXPs to work together for the responsible stewardship and sustainability of the Internet.

https://manrs.org/ixps/

IXP	Date Approved	Action 1 Prevent	Action 2 Promote	Action 3 Protect	Action 4 Coordinate	Action 5 Tools
Berlin Commercial Internet Exchange	2nd Sep 2019	✓	• 2-1: Assist (1) • 2-2: Assist (2) • 2-3: Promote	✓	✓	✓
IXP	Date Approved	Action 1 Prevent	Action 2 Promote	Action 3 Protect	Action 4 Coordinate	Action 5 Tools

Further input

Answers:

- peering works, it's time-consuming to get started and it's worth starting sooner rather than later - for a better internet
- Consider RFC8950 (IPv4 with IPv6 next-hop) for interconnection!
- We recommend to test every batch of 400G transceivers (with new firmware).
- We are always looking for more non-profit communities and sponsors of IP transit, connectivity and hardware (switches). – CommunityIX

Feedback from the IXPs

MegaIX

Highlight

Global additional IX buildouts on the Fabric

Change

Consolidation on ports, higher utilisations

Optimise

Restarted regional "beerings"

Next thing

Continue to lead with an automation first, value for money IX

KA-NIX

Highlight

We’ve received another batch of new hardware.

Change

Optimise

Next thing

A complete overhaul of the technical and administrative organization is underway. 100G ports will be available next year.

LocIX

Highlight

Established 100G Peering Ports with Cloudflare, Valve, HE, Hetzner and others.

Change

100G Equipment went live

Optimise

Next thing

100G Expansion between our POPs

Equinix IX

Highlight

250G peak traffic in March on our Frankfurt IX

Change

New feature on our portal: "TopTalkers," which leverages sFlow-based analysis. Providing insights into public traffic status for each IX, available on our public webpage.

Optimise

Easily identify Top Peers by day or hour and spot unusual traffic sources in real time with our enhanced analytics and optimizing the back-end.

Next thing

Portal: "TopTalkers"—an sFlow-based analytics feature, with real-time data available directly on our public webpage.


intel
intel
hybula

ERA-IX



Highlight

Our first expansion into Frankfurt after our success in Amsterdam.

Change

Moving away from a traditional IXP towards a fabric, by offering value added services like premium peering, transport, a marketplace, and finally preparing our backend for a fully automated portal.

Optimise

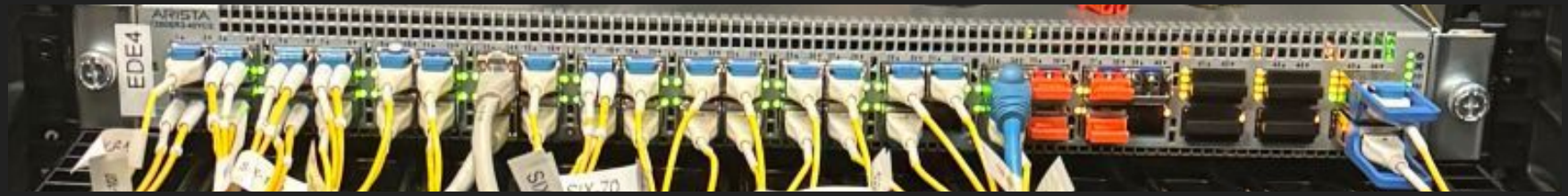
Automate the port turn-up/testing process and our next phase is automatic contract signing.

Next thing

Offering 400G ports, expanding into other metros, releasing our fabric portal.

Arista, Arista, Arista ?? - but which IXP ??

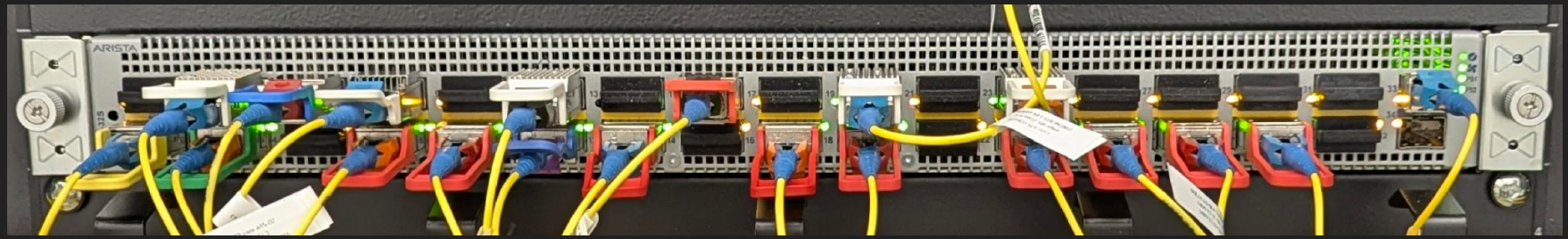
A



B



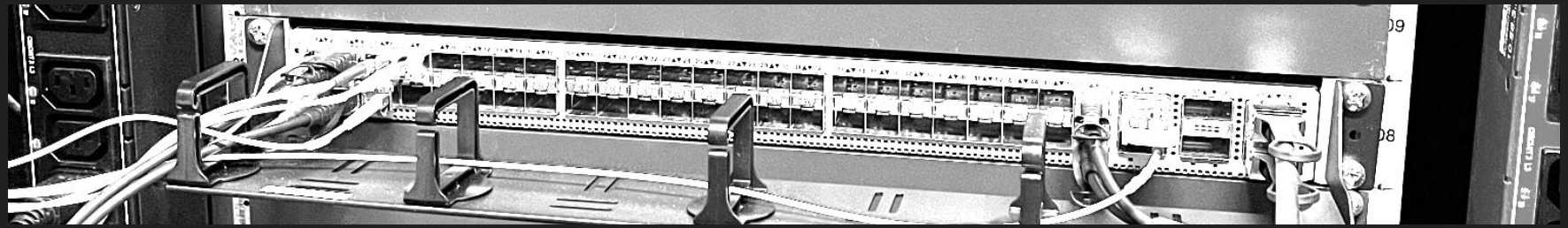
C



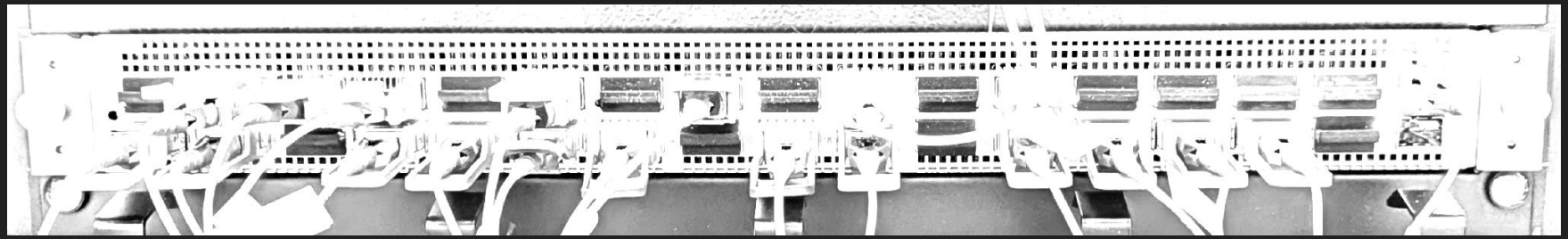
A

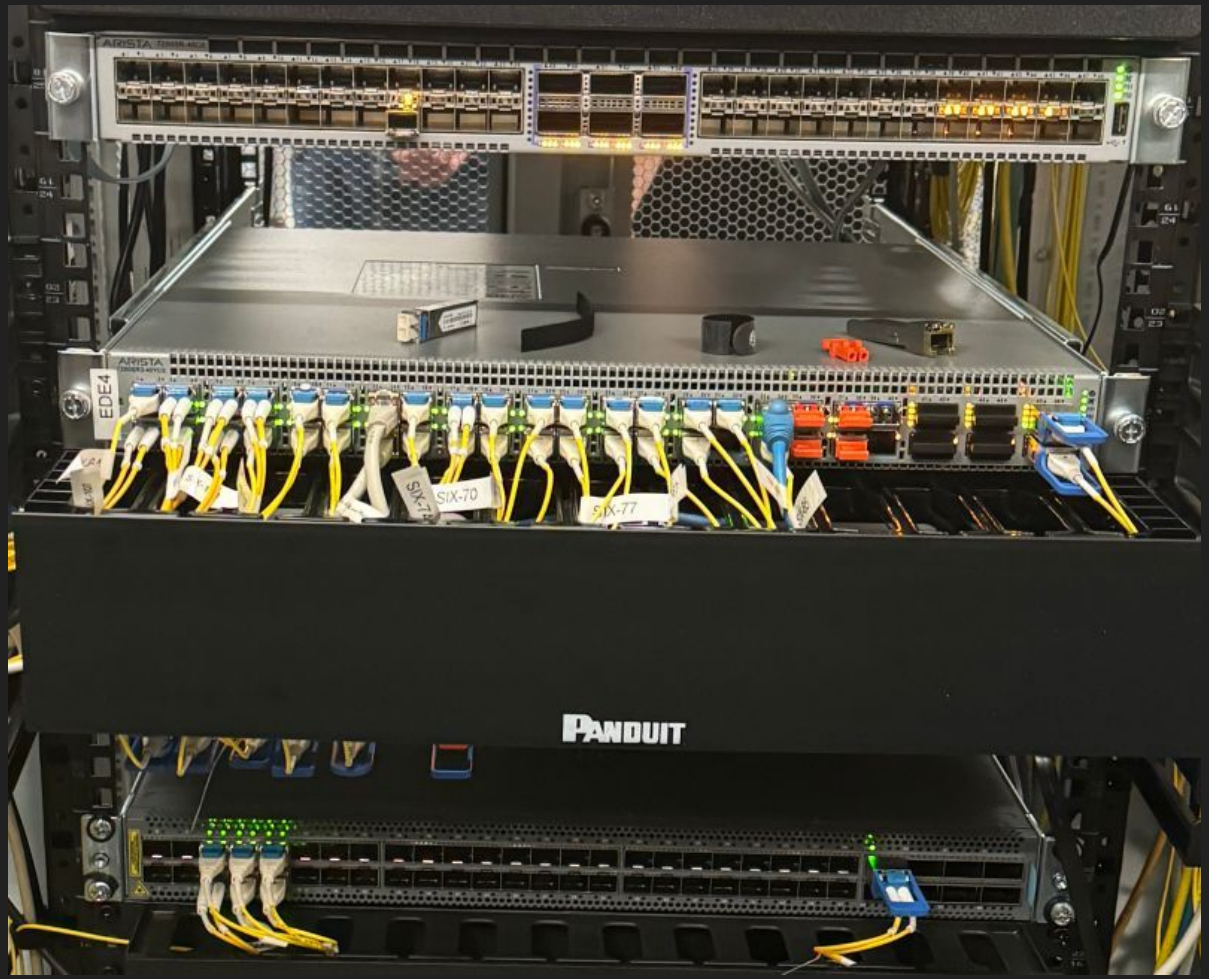


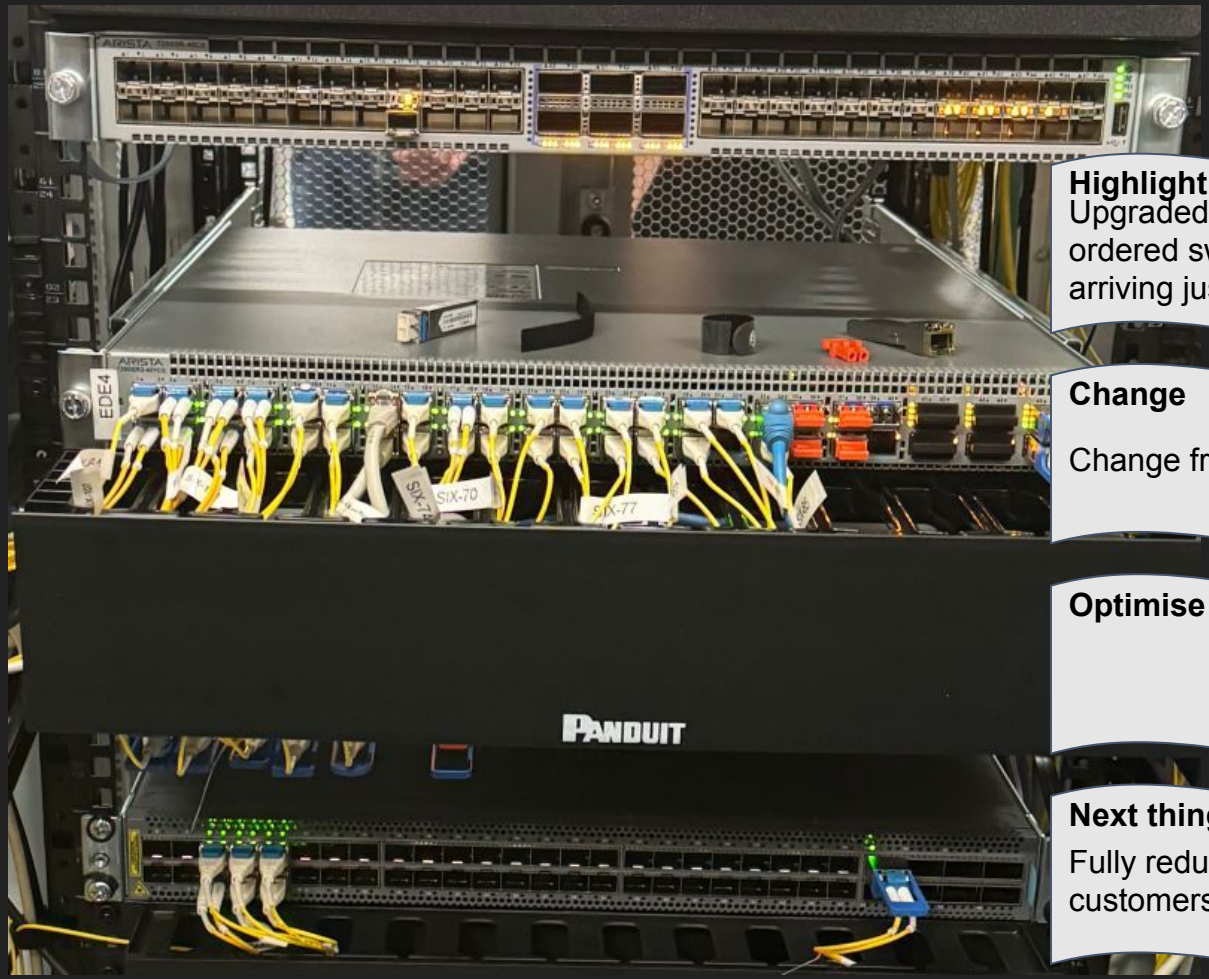
B



C




ARISTA
EDE4
SIX-70
SIX-77
PANDUIT



Stuttgart-IX

Highlight
Upgraded to fully owned and supported hardware, ordered switches for an additional site, which are arriving just now.

Change

Change from plain L2 to MPLS/EVPN with SR

Optimise

Next thing
Fully redundant fabric offering diversity to our customers

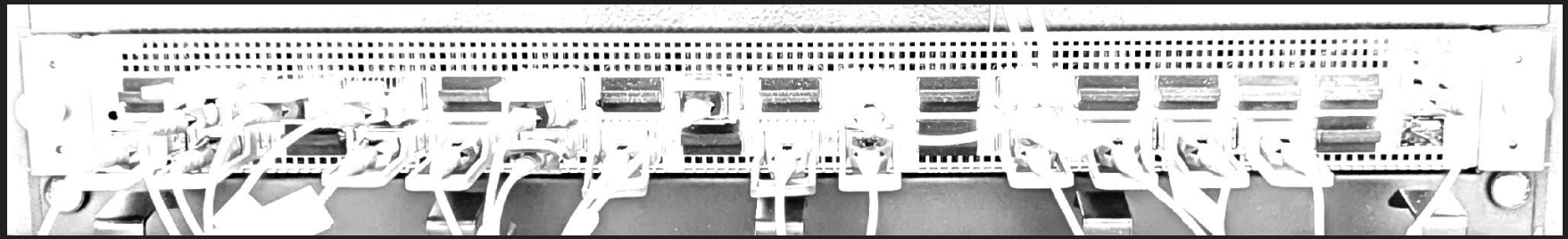
A



B



C

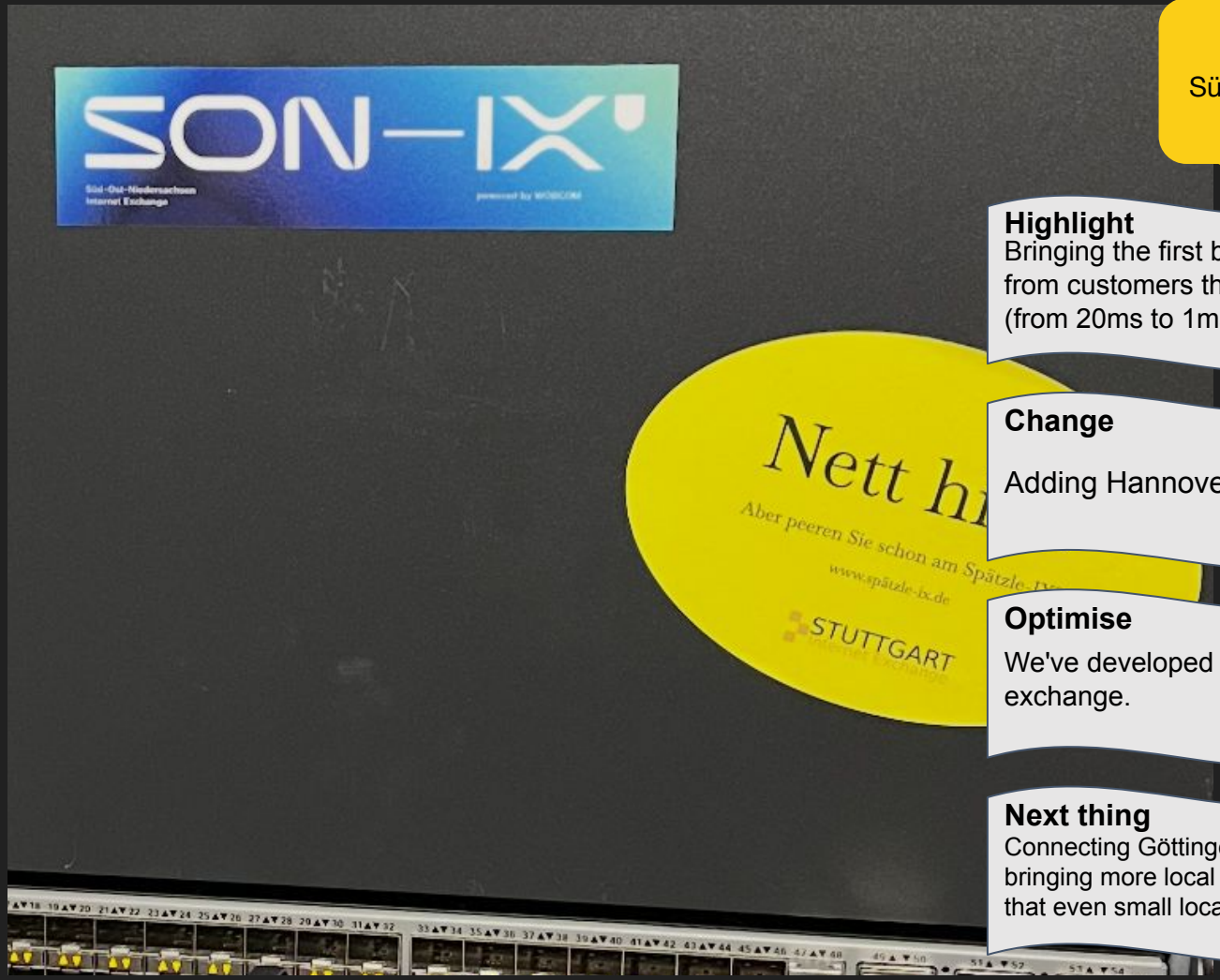



SON-IX
Nett hier.
STUTTGART



And yes,
I’m a Cisco :-)


SON-IX
Süd-Ost-Niedersachsen
Internet Exchange
Nett hier.
Aber peeren Sie schon am Spätzle-IX?
www.spätzle-ix.de
STUTTGART



SON-IX
Süd-Ost-Niedersachsen
Internet Exchange

Highlight
Bringing the first bits to the Exchange and getting feedback from customers that actually notice the difference in latency (from 20ms to 1ms).

Change

Adding Hannover as a new metro area to our network.

Optimise

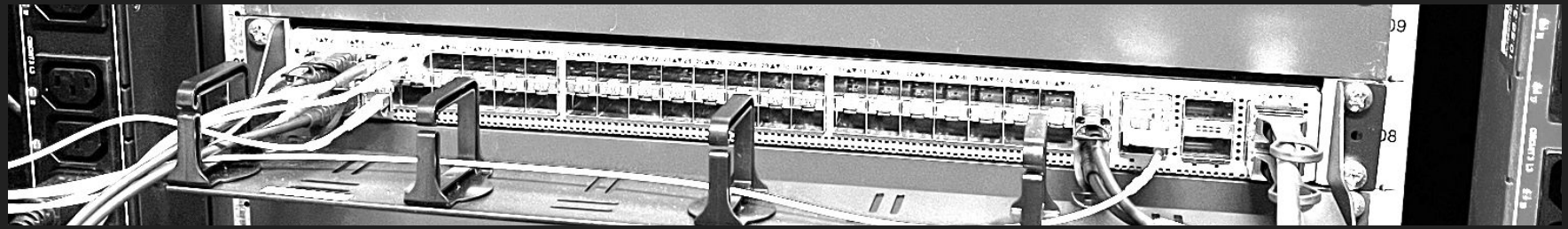
We've developed some branding and set up a website for the exchange.

Next thing
Connecting Göttingen as the next metro to our network and bringing more local networks to the exchange. Beeing an example that even small local IXPs work and can make a difference!

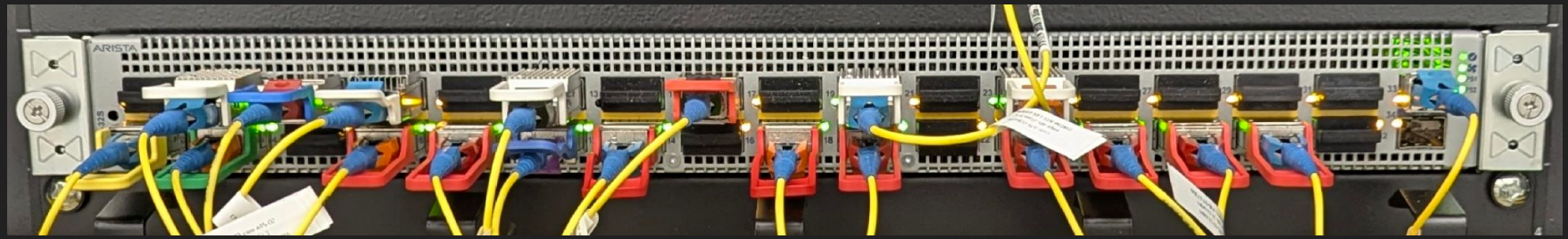
A

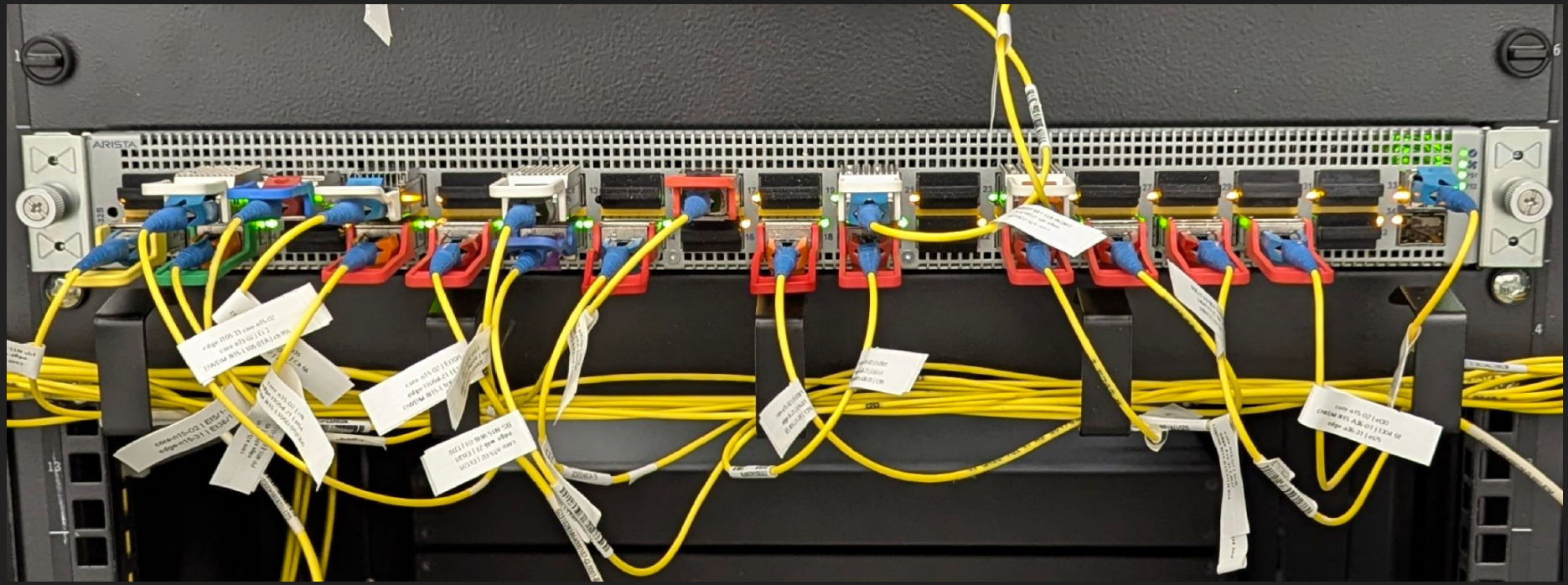


B

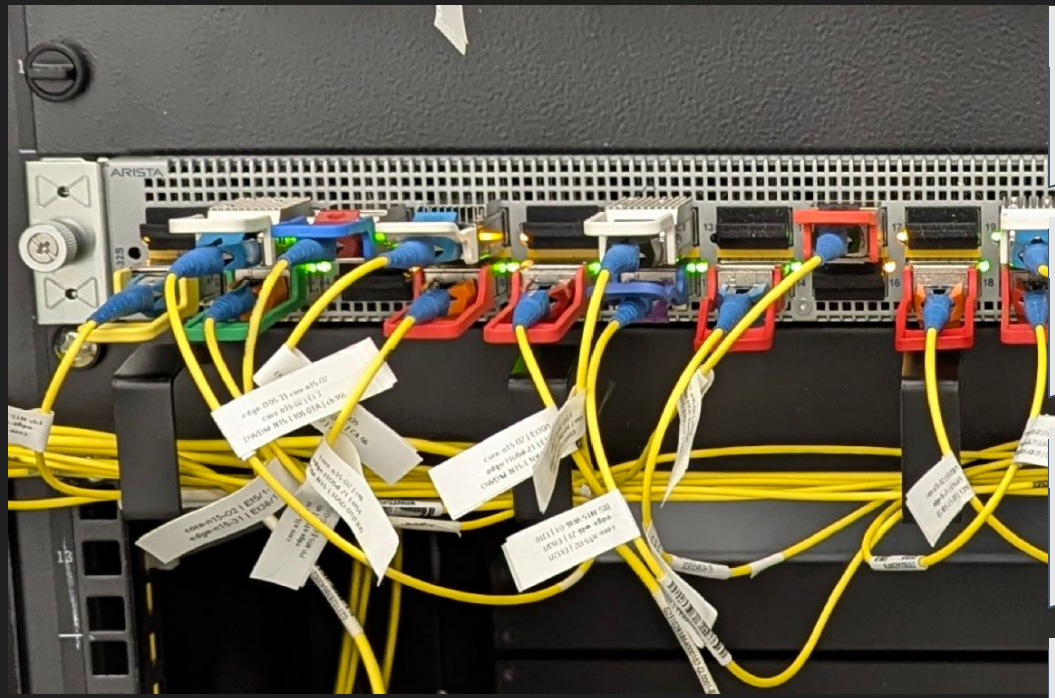


C




ARISTA

BCIX



Highlight

Upgrade of our core network using 400G without customer impact.

Change

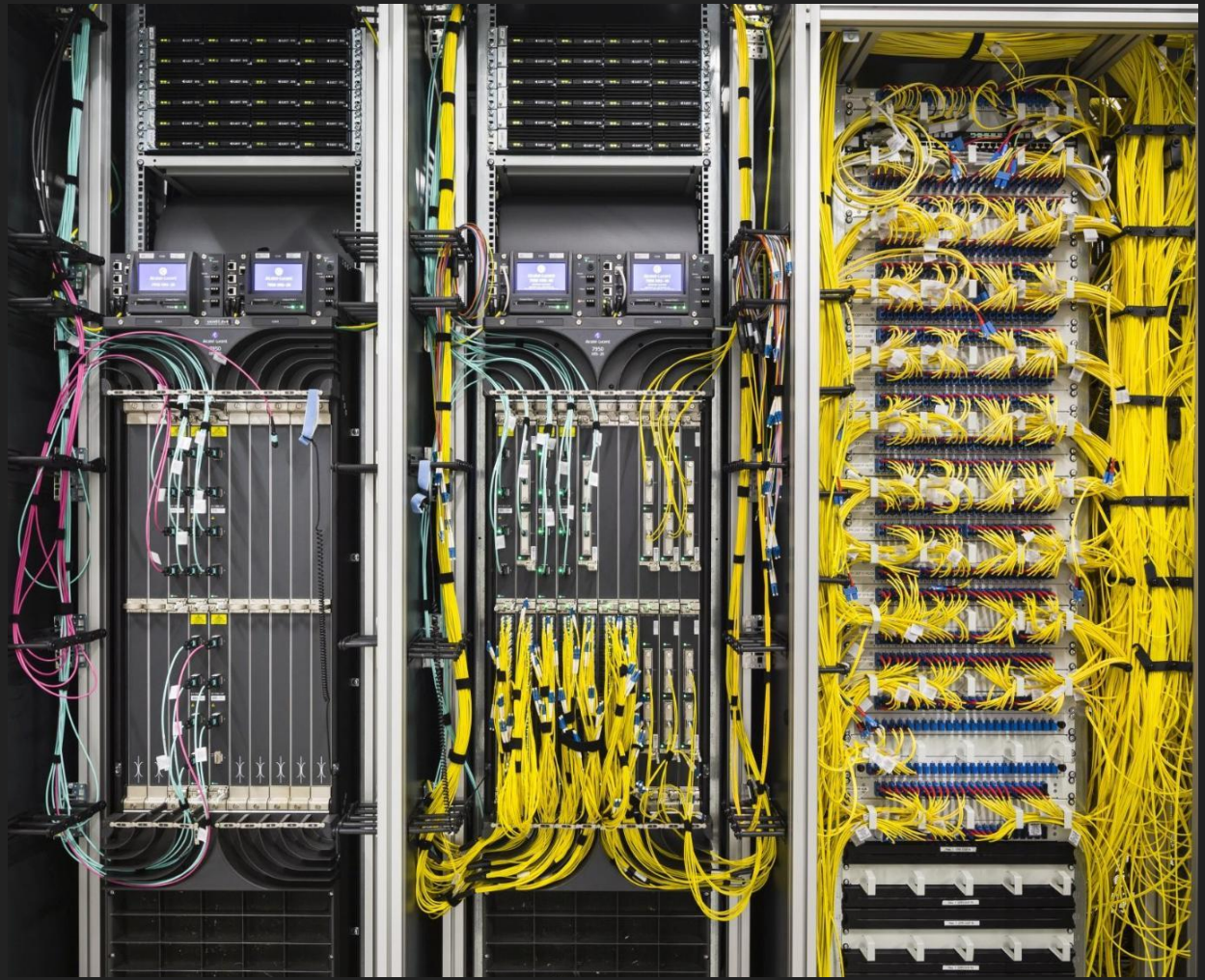
Brushing up our ISMS to achieve ISO 27001 certification.

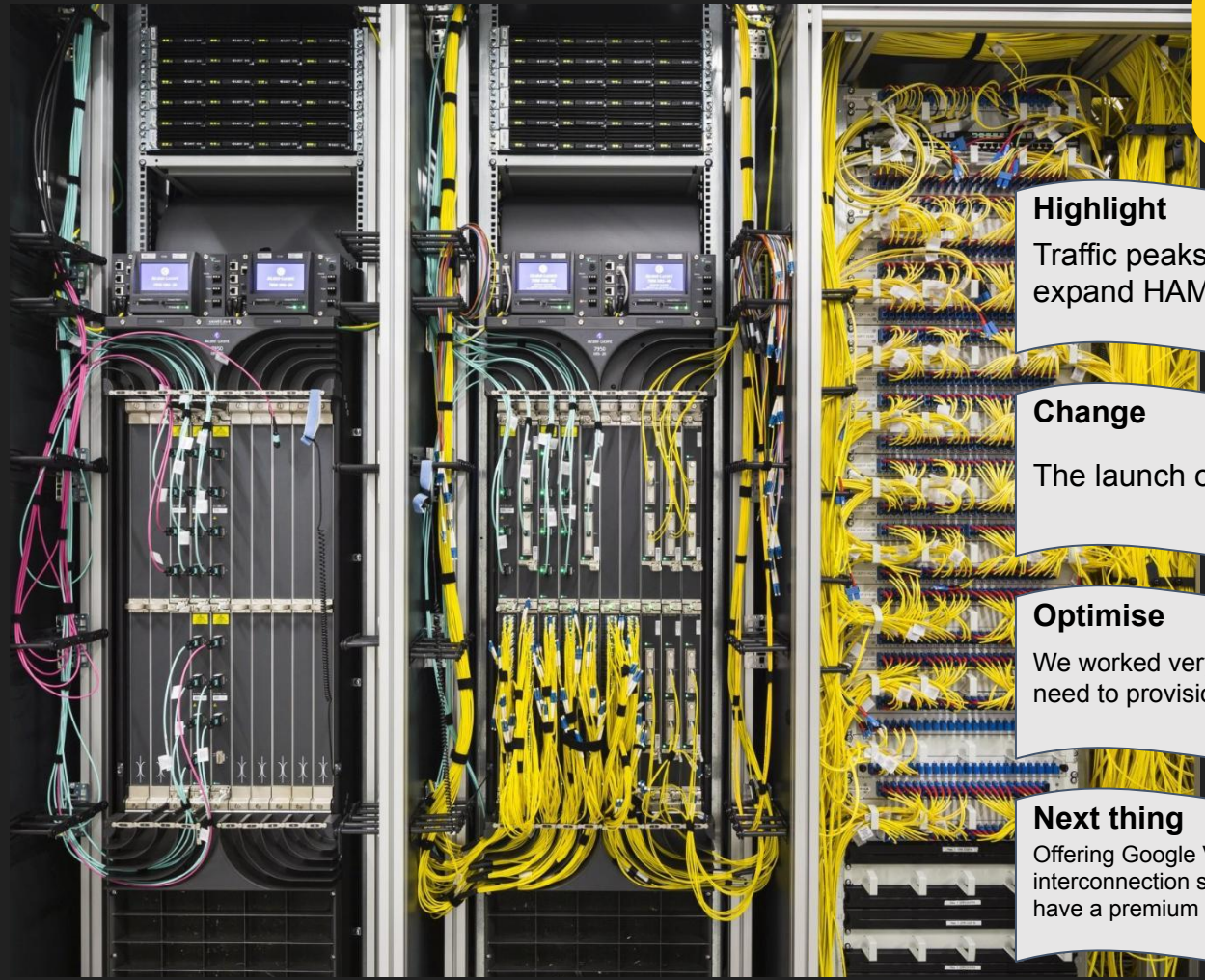
Optimise

"Beautification (and renovation) of our offices.
Automation of config generation for our main infrastructure."

Next thing

Support programs for community projects and local IXPs.





DE-CIX

Highlight

Traffic peaks FRA 17,8 Tbps + MUC 304 Gbps + expand HAM to Oldenburg with eweTEL

Change

The launch of Cloud ROUTER

Optimise

We worked very hard on optimising the time the Customer Service need to provision a newly ordered service.

Next thing

Offering Google VPP (Verified Peering Provider), MAPS and similar interconnection services to support the small-to-medium sized networks to have a premium peering with large content networks.

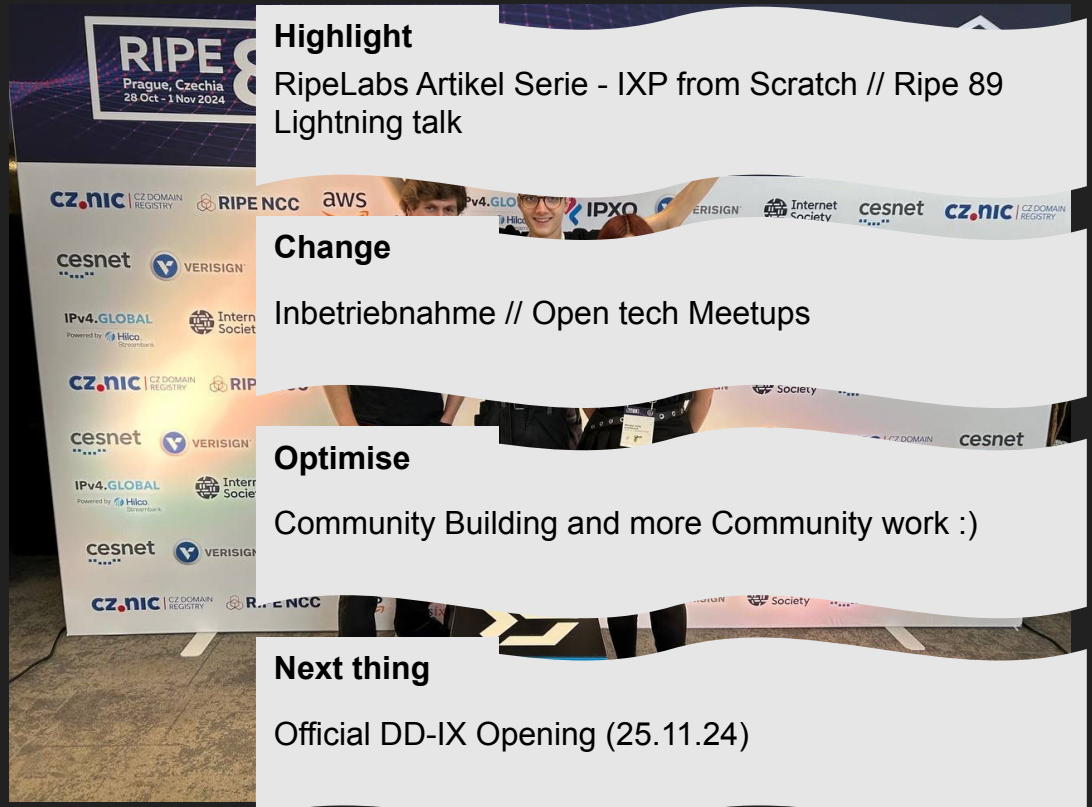

Dessert




RIPE 89
Prague, Czechia
28 Oct - 1 Nov 2024
#RIPEMeeting
WE BELIEVE IN OPEN SOURCE



DD-IX
Dresden Internet
Exchange e.V.



Highlight

RipeLabs Artikel Serie - IXP from Scratch // Ripe 89 Lightning talk

Change

Inbetriebnahme // Open tech Meetups

Optimise

Community Building and more Community work :)

Next thing

Official DD-IX Opening (25.11.24)


NLIX

NL-IX



Highlight

Reaching 10T peak traffic

Change

Extended our network to Dublin

Optimise

Significant improvements in our customer onboarding process

Next thing

Deploying anti-DDoS services that can filter on the IX LAN


der berg

Rheintal-IX



Highlight

Concerted hardware upgrade and config change at 5 sites with minimal downtime and the team event that followed.

Change

New 10G modules for peer access.

Optimise

Virtualisation platform and redundant IP feeds for our services.

Next thing

Ring around Lake of Constance and new PoP in Friedrichshafen - project will be finalised this year at last.

Any other comments?

Keep Local Traffic Local! :)

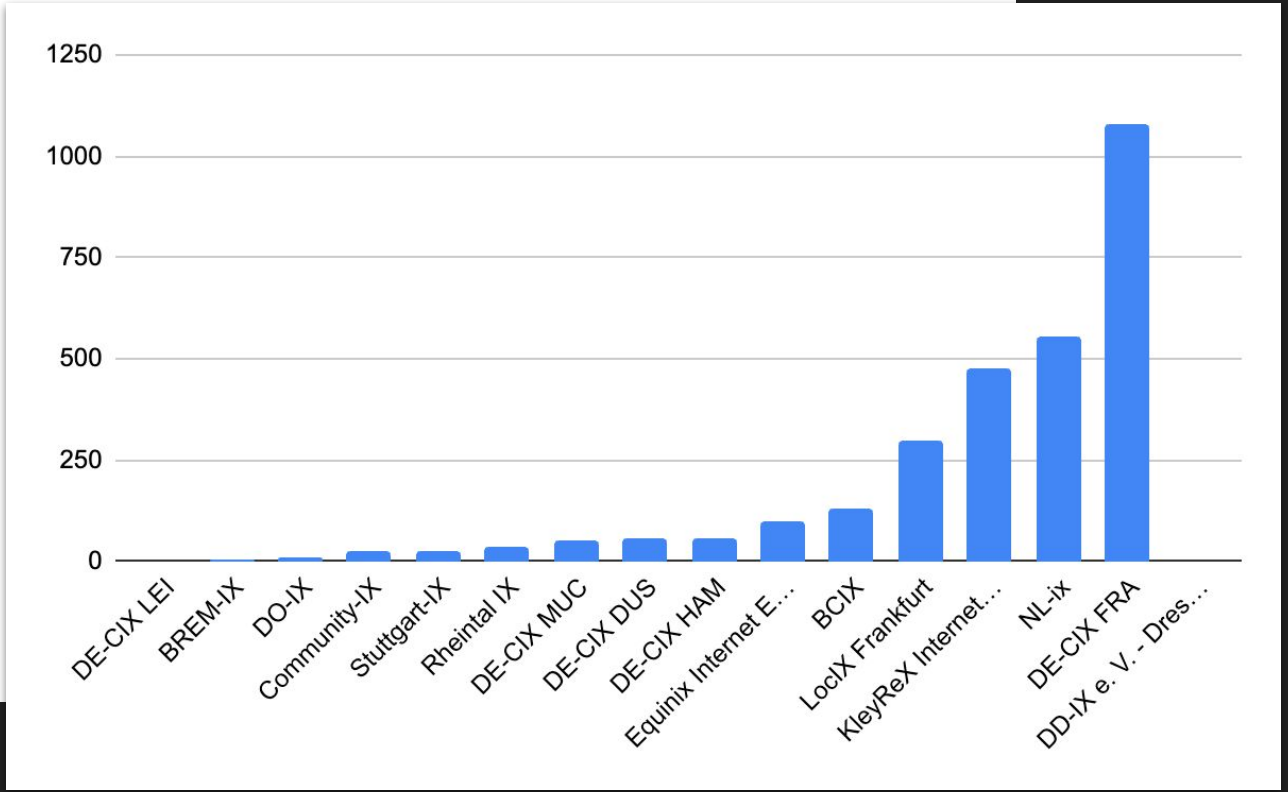
Use more bandwidth!

DENOG rocks

Thank you

Number of ASNs (per metro)

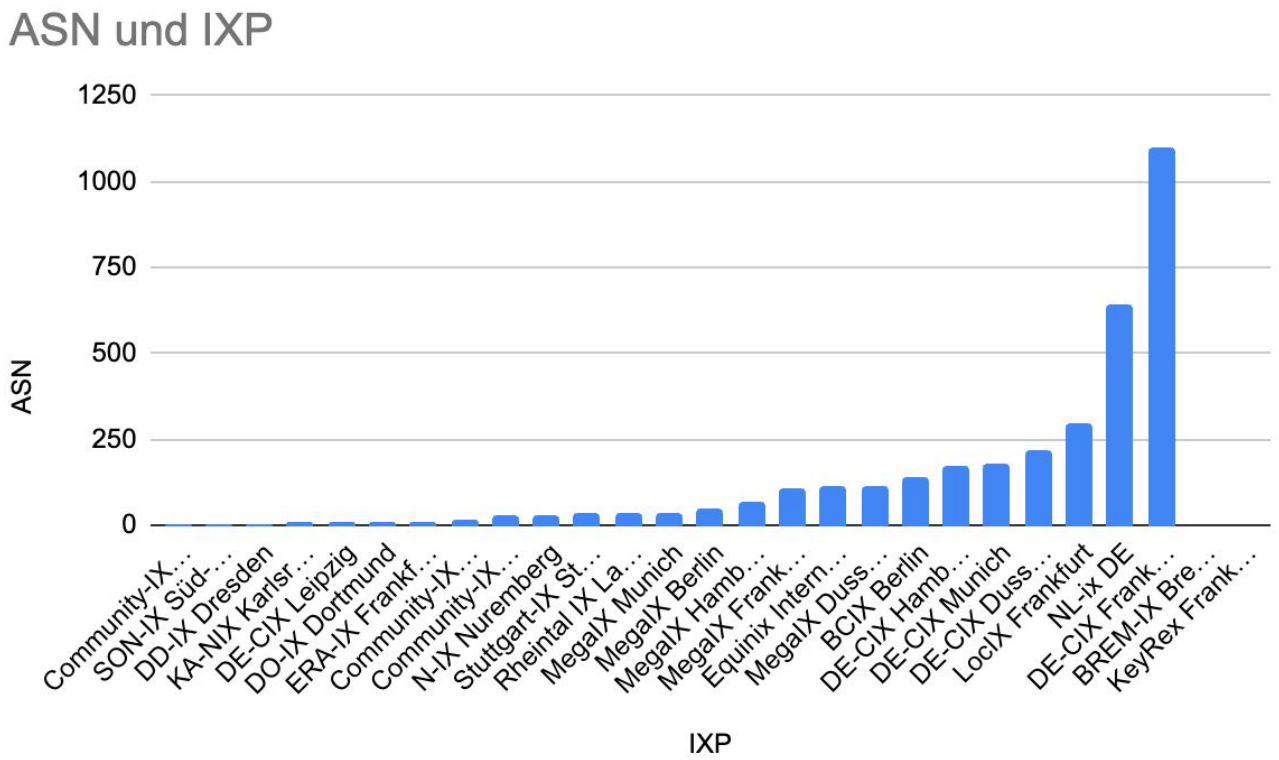
Average: 195



Number of ASNs (per metro)

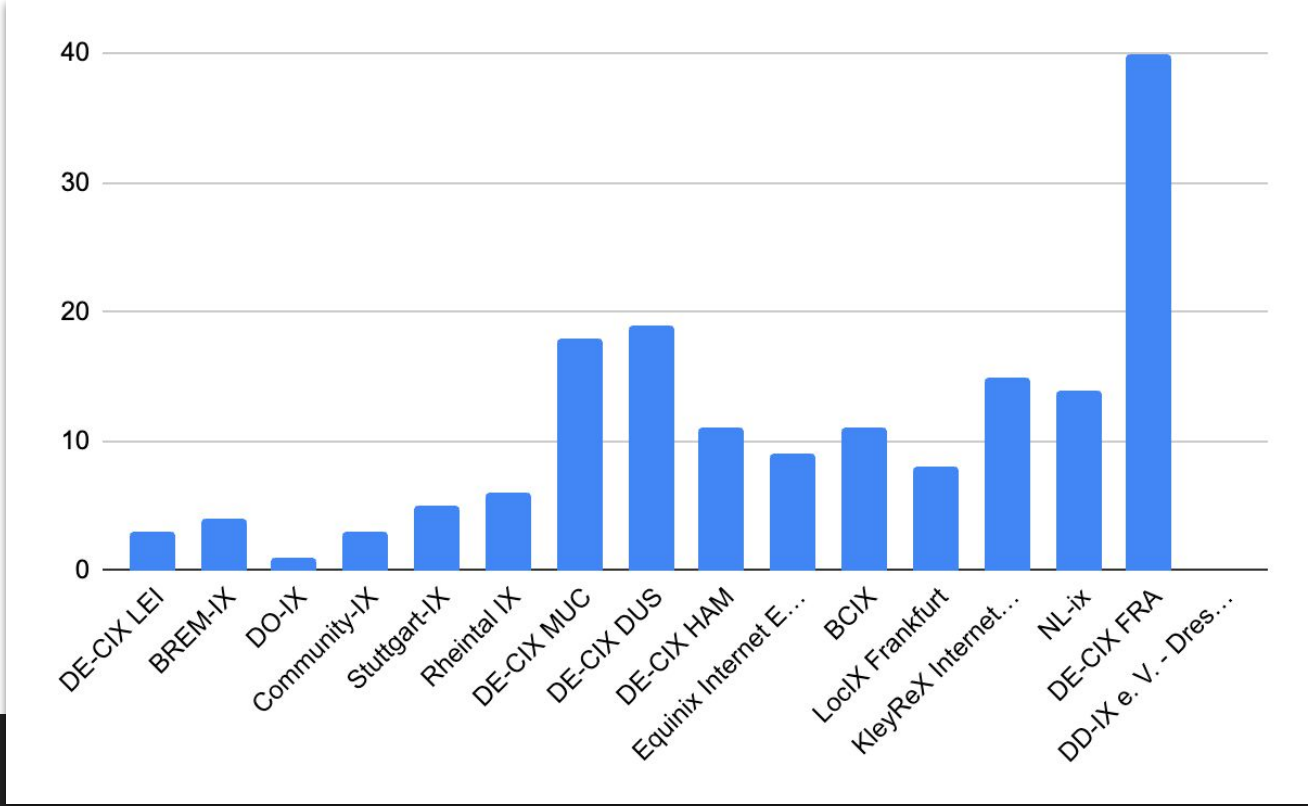
Average: 138
Last year: 183

	2024	2023
Avg	138	183
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Min	4	0



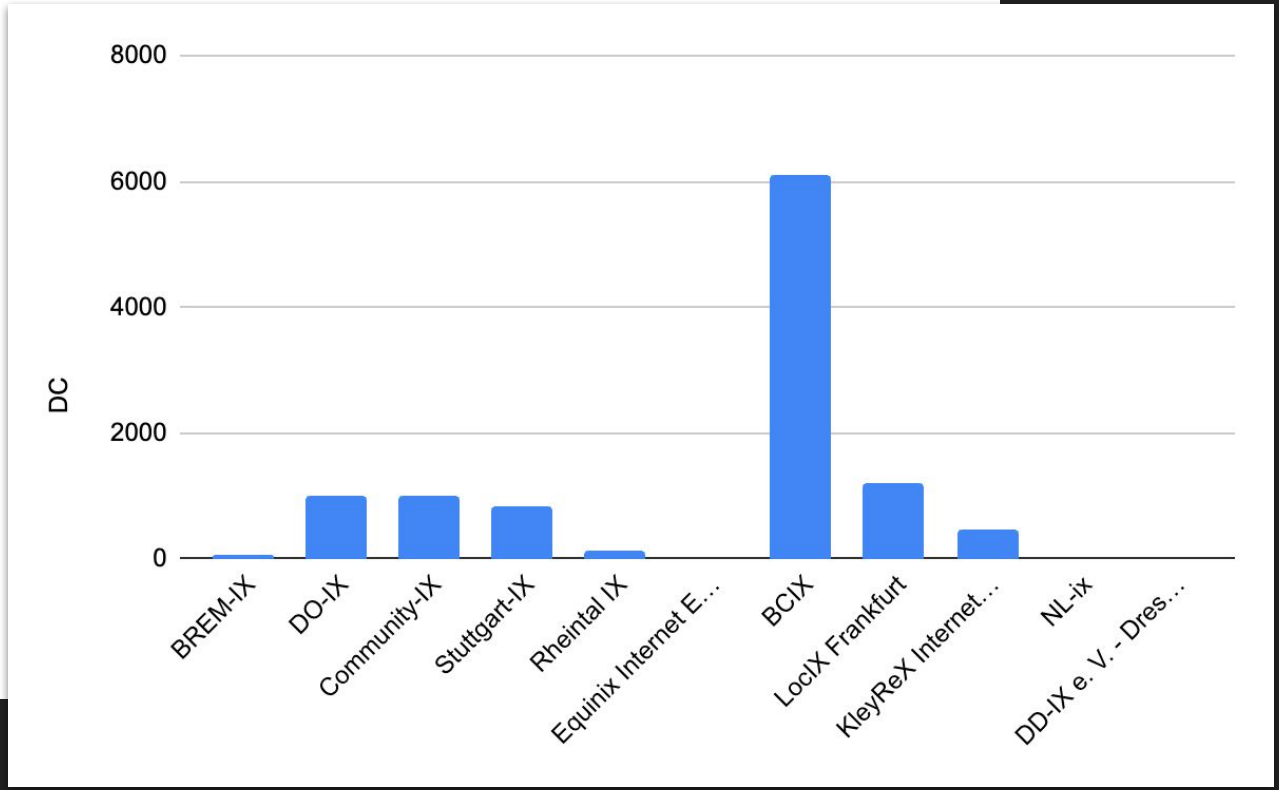
Number of sites (per metro)

Average: 11



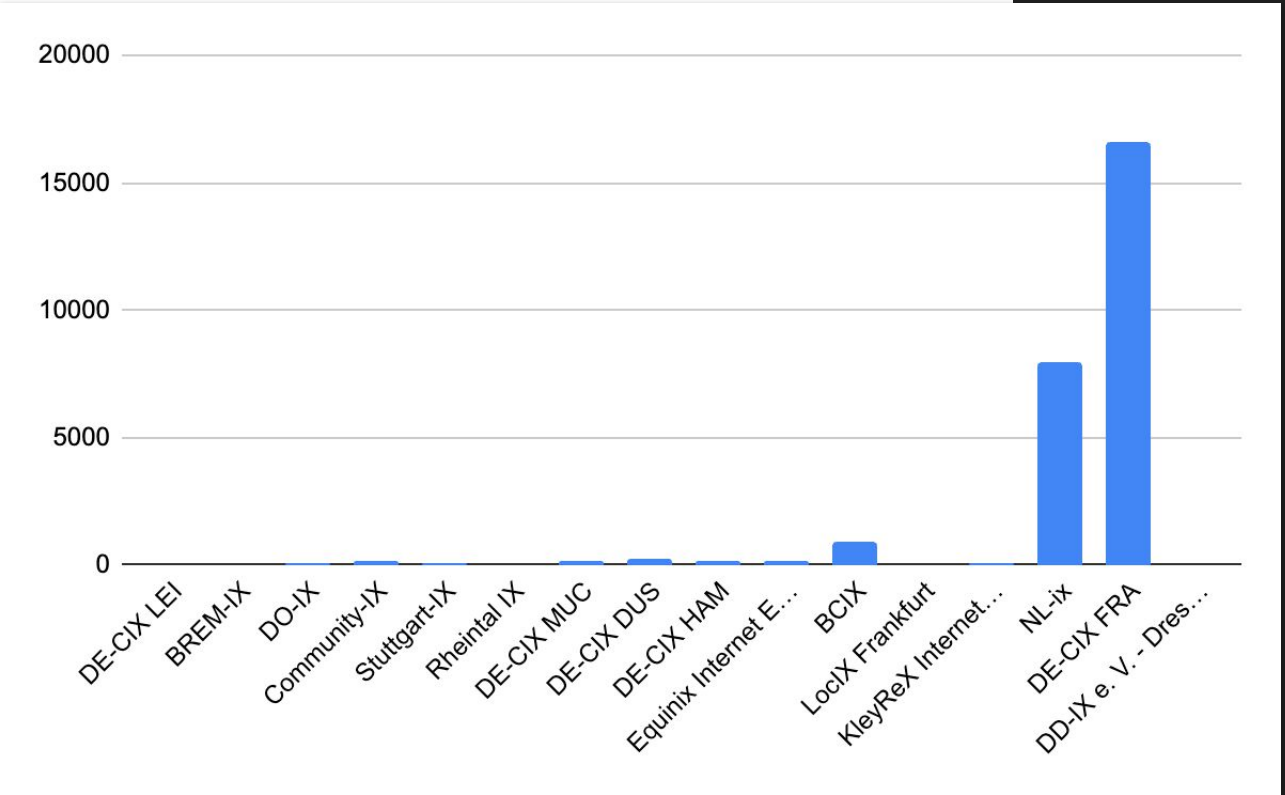
Connected Capacity

Average: 1352G
(8x n/a)



Peak traffic

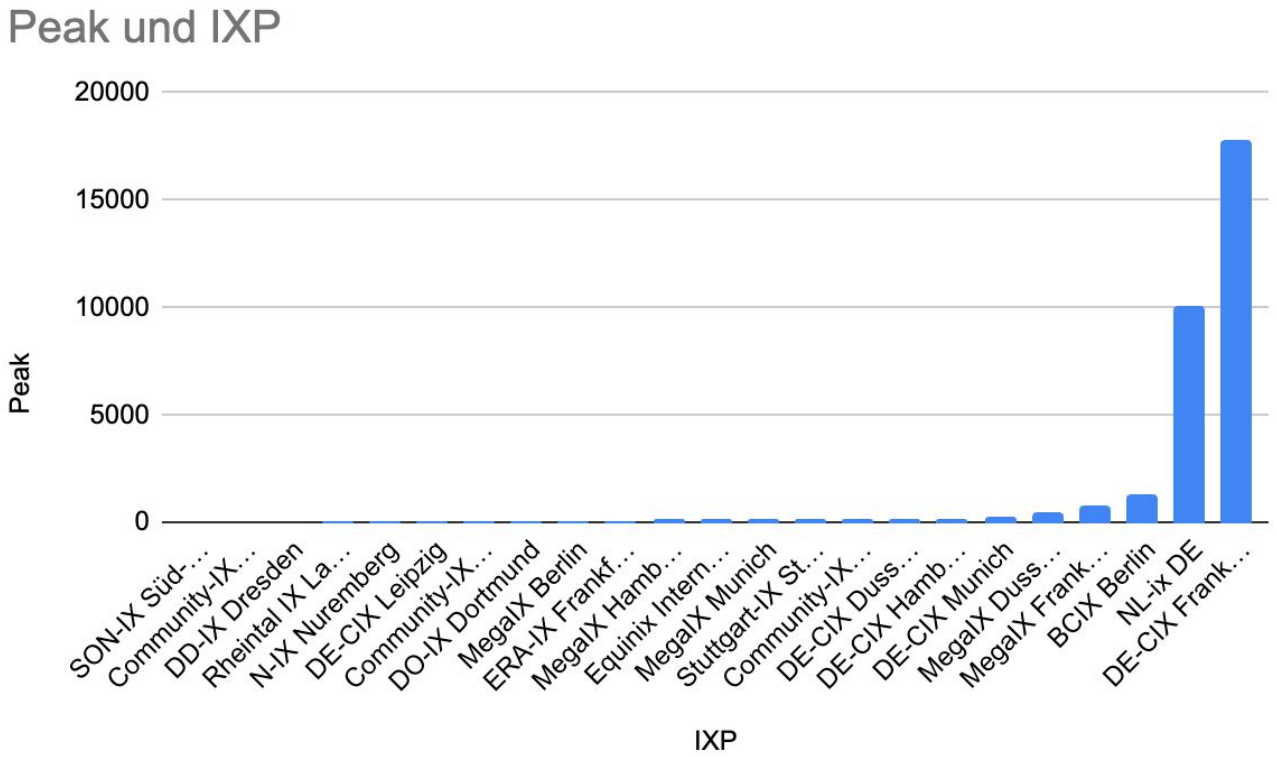
Average: 1905G



Peak traffic

Average: 1.390G
Last year: 1.778G

	2024	2023
Avg	1.390	1.778
Max	17.800	16.620
Min	0,1	0



Peak traffic (without DE-CIX FRA and NL-IX)

Average: 197G

	2024	w/o DE-CIX FRA and NL-IX
Avg	1.390	197
Max	17.800	1.285
Min	0,1	0,1

